

1. A certain hybrid car uses 39 gallons of gasoline to travel 1,950 miles. At the rate, how many kilometers does the car travel using 39 gallons of gasoline?

(Use 1 mile = 1.6 kilometers)

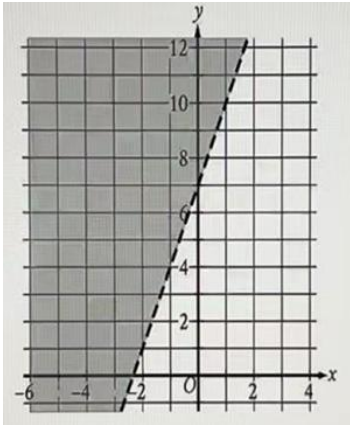
- A. 312
B. 3,120
C. 31,200
D. 312,000
2. The function h is defined by $h(x) = 4x - 9$. What is the value of $h(-3)$?
- A. 21
B. 13
C. -13
D. -21
3. The solution to the given system of equations is (x, y) . What is the value of y ?
- $$10x = 110$$
- $$8x - 86 = y$$
- A. 86
B. 11
C. 10
D. 2
4. Which expression is equivalent to $7x^5 - 6x^4 + 9x^3$?
- A. $x^4(7x - 6)$
B. $x^3(7x^2 - 6x + 9)$
C. $9x^3(7x^2 - 6x + 1)$
D. $6x^5(-6x^4 + 9x^3 + 1)$

5. Which ordered pair (x, y) is a solution to the given system of equations?

$$\begin{aligned}y &= 0 \\ y &= 8(x^2 - 36)\end{aligned}$$

- A. (0,6)
B. (0,36)
C. (6,0)
D. (36,0)
6. A group of students purchase ice cream cones and ice cream sundaes. The cones cost \$1.40 each, and the cost for 19 sundaes is \$64.60. The total cost for the cones and sundaes is \$85.60. Which equation represents this situation, where x is the number of cones purchased and y is the number of sundaes purchased?
- A. $1.40x + 64.60y = 85.60$
B. $64.60x + 1.40y = 85.60$
C. $3.40x + 1.40y = 85.60$
D. $1.40x + 3.40y = 85.60$
7. Hakim has a bag of 50 tiles, where 45 are labeled with a letter and 5 are blank. If Hakim selects a tile from the bag at random, what is the probability of selecting a blank tile?
8. In triangle DEF, the measure of $\angle D$ is 114° . Which of the following could be the measure, in degrees, of $\angle E$?
- A. 115
B. 90
C. 67
D. 65

9. The shaded region shown represents the solution to which inequality?



- A. $y > 3x + 7$
B. $y > 3x - 7$
C. $y < 3x + 7$
D. $y < 3x - 7$
10. Which expression is equivalent to $20x^2 + 3 + 5x^2$?
- A. $25x^4 + 3$
B. $25x^2 + 3$
C. $28x^4$
D. $28x^2$
11. A company opens an account with an initial balance of \$35,600.00. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A , where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 15 years is \$129,672.38. Which equation could define A ?
- A. $A(t) = 35,600.00(1.09)^t$
B. $A(t) = 35,600.00(0.09)^t$
C. $A(t) = 94,072.38(1.09)^t$
D. $A(t) = 94,072.38(0.09)^t$

12. What is the y –intercept of the graph of $y = 14^x$ in the xy -plane?

- A. (0,15)
- B. (0,14)
- C. (0,1)
- D. (0,-1)

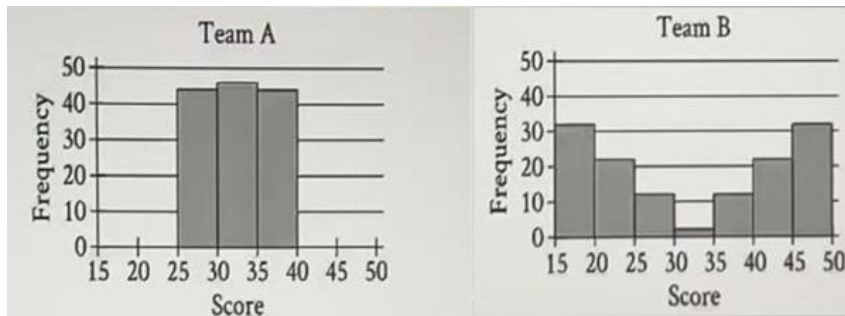
13. The function f is defined by $f(x) = 2x + b$, where b is a constant and $f(7) = 6$.

What is the value of b ?

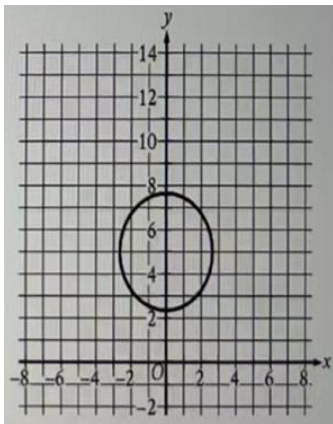
14. A number f is 120% greater than a positive number g . A number h is 90% less than the number f . The number h is how many times the number g ?

- A. 0.12
- B. 0.22
- C. 1.08
- D. 1.98

15. The histograms summarize the distribution of scores for team A and team B. Which statement best compares the standard deviation of scores for these teams?



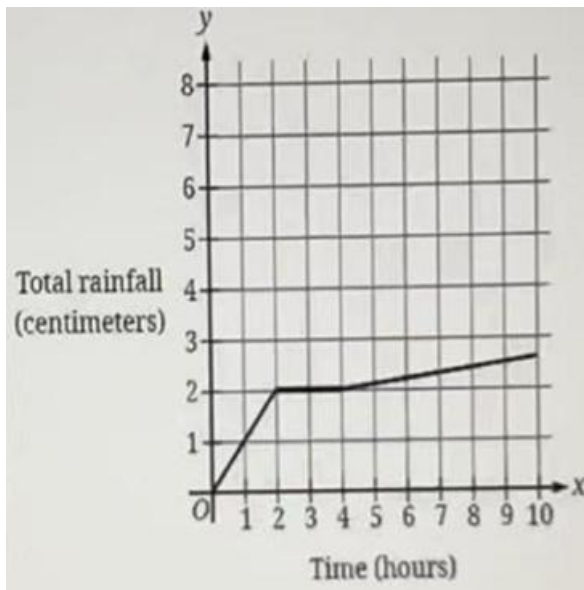
- A. The standard deviation of scores for team A is less than the standard deviation of scores for team B.
- B. The standard deviation of scores for team A is greater than the standard deviation of scores for team B.
- C. The standard deviation of scores for team A is equal to the standard deviation of scores for team B.
- D. There is not enough information to compare the standard deviations.
16. Circle A shown is defined by the equation $x^2 + (y - 5)^2 = 7$. Circle B (not shown) has the same radius but is translated 91 units to the right. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, where h, k , and a are constants, what is the value of $4a$?



17. The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

$$h(t) = -16t^2 + b$$

18. The graph shows the total amount of rainfall y , in centimeters, from the start of a 10-hour period, where x is the number of hours after the start of the period. Which of the following statements about the rainfall during this time period is true?
- A. The rate of rainfall was 2 centimeters per hour between $x=2$ and $x=4$.
 - B. The rate of rainfall was the greatest between $x=4$ and $x=10$.
 - C. The rate of rainfall increased between $x=0$ and $x=2$.
 - D. The rate of rainfall was 0 centimeters per hour between $x=2$ and $x=4$.



19. For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48. What is the value of the greatest of the three integers?

20. A rectangular banner has an area of 2,300 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

- A. 2,320
- B. 2,340
- C. 2,760
- D. 3,312

21. What is the sum of the solutions to the given equation?

$$(x - 42)^2 = 1$$

- A. 42
- B. 43
- C. 84
- D. 85

22. In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k ?

$$5(kx - n) = -\frac{65}{14}x - \frac{85}{18}$$